

Typed Wall Morphology for Sectorized Observable Frameworks

Sparse Wall Records, Coordinate Profiles, Multi-Label Taxonomy, and Local-Model Eligibility

WuJun Chen
Independent Researcher
2026

This paper is Paper XI of the RIME program. It develops wall-record morphology, profile-relative coordinates, multi-label curation, and local-model eligibility over the typed interfaces of Papers VIII–X. Strict records require upstream wall admission; analogue morphology does not.

Abstract

Problem. A typed wall and a source-addressed finding do not yet determine how the observed change should be recorded or organized. Fixed vectors such as $(\Delta R_1, \Delta R_2, \Delta D, \Delta \tau, \Delta P_d)$ conflate distinct carriers, force absent coordinates to look like zeros, and can promote static findings or trajectory diagnostics into wall events.

Approach. For the strict branch, starting from an admissible wall datum supplied by the typed deformation interface, this paper defines an ordinary record construction $\text{RecordWall}_{P_W}(\mathfrak{D}_\kappa)$. A separate analogue construction records included morphology without claiming wall admission. The corpus separates morphology-record bundles from non-wall context records and distinguishes strict wall records from included analogue morphology. Each morphology bundle contains atoms that are either oriented trajectory events with sparse before/after maps or domain-level locus samples with incident-stratum germs. In the strict branch, one primary field carries the record’s upstream wall admission; in the analogue branch, it anchors morphology without asserting strict admission. Separately registered context fields remain co-observations. A Wall Profile P_W selects fields and policies, while six structured coordinate families record morphology and evidence. The declared mechanism labels serve only as nonexclusive curation tags. A separate eligibility gate determines whether a sufficiently smooth local branch may be compared with an ADE-type model.

Results. The resulting architecture is

Typed Wall Records $\longrightarrow$ Wall Coordinate Profiles
 $\longrightarrow$ Multi-Label Taxonomy
 $\longrightarrow$ Local-Model Eligibility.

The finite census reads a source-pinned 28-row census, of which 27 rows remain active. A versioned corpus-inclusion and upstream-admission reference ledger places 5 strict wall bundles in the strict wall spectrum and retains 2 analogue morphology bundles in a separate analogue morphology set. The remaining active rows are 5 static boundary witnesses, 2 pre-wall references, and 13 trajectory diagnostics. The 27 active records carry 34 recomputed nonexclusive curation-tag memberships. Three anchors exhibit the intended range: spectral pair-gap morphology with a branch-aware $A_2 \rightarrow A_1 + A_1$ falsification audit, a paired affine/unitary CNOT path-admissibility audit excluded from wall admission, and a discrete graph endpoint.

Boundary. The construction is not a functor and no category of deformation records is asserted. Wall coordinates are relative to the declared carrier, chart, path, threshold, cutoff,

normalization, sampling, and profile. Static repair findings, plateau intervals, and rate separation are not walls without a declared change locus. The census is not a complete classification, and ADE terminology is restricted to eligible smooth local-model candidates.

Notation Table

Symbol	Meaning
$\mathcal{F} = (V, Q, Y; X, \mathcal{H})$	typed Sectorized Observable Framework (SOF), with optional independently declared Lie/Hall data
$\mathfrak{D}_\kappa$	admissible wall datum for a selected typed field κ
P_W	Wall Profile declaring fields, conventions, and policies
$\text{RecordWall}_{P_W}(\mathfrak{D}_\kappa)$	ordinary wall-record construction
e	one morphology atom
WallCorpusEntry	disjoint union of morphology-record bundles and wall context records; morphology bundles split into strict wall and analogue morphology records
$\mathcal{C}$	a finite corpus of WallCorpusEntry values
MorphologyAtom	disjoint union of trajectory events and locus samples
K_e	finite set containing the primary field and any registered context fields
δ_e^γ	oriented sparse before/after map for a trajectory event
δ_e^{loc}	incident-stratum germ for a locus sample
$\text{Prof}_W^{P_W}(e)$	structured wall coordinate profile
$\text{MorphSig}_W^{P_W}(e)$	curation-independent morphology signature
$\text{CuratedSig}_W^{P_W, v}(e)$	versioned pairing of a morphology signature with a curation assignment
$\text{Spec}_{W, \text{str}}^{P_W}(\mathcal{F}, \mathcal{C}; \chi)$	profile-relative multiset of admitted strict-SOF morphology signatures in context χ
A_Y^+	positive-word algebra $\text{alg}_{\mathbb{C}}(I, Y)$
A_Y^*	observable star-closure $C^*(Y)$
$A_{Q, Y}^*$	sector-enriched star-closure $C^*(D_Q \cup Y)$
$R_1[Y]$	aggregate direct operator support
$\text{Route}_d[Y], W_d[Y]$	routed-product and full-word support at length d
$R_1^{\text{Lie}}, R_2^{\text{Lie}}$	direct and simple-commutator support on a registered Lie/Hall carrier
D_κ	exact first-hit depth, when certified
$D_\kappa^{(\leq d_{\max})}$	truncated first-hit depth with an explicit cutoff
$\text{UNREACHED_AT_CUTOFF}$	finite-audit state; never mathematical infinity
A_k, D_k, E_k	Arnold singularity families, used only after local-model eligibility

Arrows in diagrams denote construction or audit order unless a theorem explicitly states otherwise.

1 Introduction

The static object language supplies marked sectors, labelled observables, distinct closure layers, and separate operator/word and Lie/Hall branches [8]. The deformation language supplies typed charts, comparison maps, trajectories, selected fields, discriminants, walls, and policy-relative response quantities [7]. Capability declarations and evidence routing are supplied by the compilation and Registry interface [3]. None of those layers needs a universal wall taxonomy.

This paper addresses the next, narrower question: once a typed wall datum or a wall-related finding is available, what record role does it have, which coordinates may be extracted, and which mechanism and regularity labels may be attached without conflating carriers?

Paper IX determines when a typed observable change qualifies as a wall. This paper does not redefine that wall; it records its typed changes, assigns a profile-relative morphology, and organizes eligible records into a multi-label taxonomy. Paper X supplies capability and evidence guards. This paper applies those guards to source-addressed records without rebuilding either upstream interface.

The stable division of responsibility is:

Paper	Owning object
VIII	static carriers, closures, and filtrations
IX	charts, trajectories, walls, pullbacks, and response policies
X	capability/evidence guards and Registry findings
XI	wall morphology, coordinate profiles, taxonomy, and local-model eligibility
XII	single-report protocol and presentation
XIII	two-report alignment and comparison

The input to this paper is therefore not a bare static SOF. A wall depends at least on a declared realization, a typed chart, a selected field, a policy, and a parameter domain or trajectory. Moreover, the deformation-record interface is not a category [7]. These facts rule out a pseudo-functor

$$\text{SOF}_{\text{str}} \longrightarrow \text{WallRecord}.$$

The construction introduced below is deliberately ordinary and profile-relative.

2 Related Work and Novelty Boundary

Program interfaces. Paper VII supplies static image–kernel incidence geometry and promotion limits [5]. Papers VIII–X supply the static SOF object language, typed deformation and wall data, and capability-aware evidence guards used here [8, 7, 3]. Downstream, Paper XII defines the single-report protocol [9], while Paper XIII defines aligned cross-report comparison [4]; neither supplies upstream wall evidence.

Local-model background. Relevant external background includes Arnold’s catastrophe and singularity theory [1, 2], stable mappings [15], stratified Morse theory [16], piecewise-smooth systems [11], and discontinuous differential equations [13].

Domain precedents. Spectral and finite-state diagnostics are compared with spectral perturbation theory [17], spectral graph theory [10], and Markov perturbation theory [19]. Additional bounded observations use Erdos–Renyi random-graph connectivity [12], finite Kuramoto dynamics [18], and Gillespie simulation [14] as domain precedents rather than SOF taxonomy theorems.

Novelty boundary. This paper begins with wall data admitted under the upstream deformation and evidence interfaces. Its contribution is the sparse record structure, profile-relative morphology, nonexclusive curation, and local-model eligibility layer. It does not redefine static SOF objects, wall admissibility, compiler emission, report serialization, cross-report alignment, or any downstream comparison result.

3 Admissible Wall Data

3.1 Upstream Input

Definition 3.1 (Admissible Wall Datum).

An admissible wall datum $\mathfrak{D}_\kappa$ consists of the data needed to invoke one declared wall notion from the deformation interface:

1. a typed local chart with fixed labels, ranks, alphabet conventions, and comparison data;
2. a selected carrier-qualified field O_κ ;
3. its declared discriminant or threshold policy;
4. a parameter domain or a selected trajectory γ ;
5. a source-addressed evidence record and claim status;
6. any cutoff, saturation, norm, normalization, censoring, or sampling policy used by that field.

Changing labels, ranks, alphabets, or filtration schemas is a chart or schema transition unless an additional comparison construction has been supplied. It is not silently recorded as a fixed-chart wall.

A wall event or locus is recorded only after its corresponding upstream admissibility conditions have been satisfied.

3.2 Realization Kind, Record Role, and Field Family

Three independent fields prevent admission, function, and mathematical carrier from being conflated:

Field	Controlled values	Function
realization_kind	strict_sof, diagnostic_analogue	states whether the source instantiates the strict SOF interface
record_role	wall_event, wall_locus_sample, pre_wall_reference, static_boundary_witness, trajectory_diagnostic, retired_provenance	states what the record does
field_family	spectral, operator, route, word, Lie-Hall, closure, state, graph, stochastic, proxy, or another declared family	identifies the typed field being observed

Every source row receives exactly one realization kind and record role. Every primary or context field receives its own field family and carrier identifier. Only upstream-admitted, corpus-included `strict_sof` records with role `wall_event` or `wall_locus_sample` enter the strict wall spectrum. A corpus-included `diagnostic_analogue` may retain event or locus morphology, but it remains in a separate analogue morphology set. Thus `diagnostic_analogue` is neither a record role nor a field family.

This gate enforces

$$\text{static repair finding} \neq \text{repair wall}.$$

A plateau interval is a regime; its onset, exit, or collapse may be a wall event. Rate separation is a trajectory diagnostic; a declared response-order crossing may be a wall event. A cutoff-unreached pair is a state; a typed transition into or out of that state may be an event.

4 Typed Wall Records

4.1 Wall Profile

Definition 4.1 (Wall Profile).

A Wall Profile P_W declares the interface used for one wall-record construction. It contains:

1. enabled typed field keys;
2. carrier, alphabet, adjoint, route, word, and Hall conventions;
3. exact or truncated depth semantics and any cutoff;
4. chart, path, orientation, and event-segmentation rules;
5. norm, threshold, normalization, censoring, and sampling policies;
6. the evidence requirements and references to any applicable upstream promotion permissions for each retained coordinate.

Missing fields are omitted or marked unavailable. They are never manufactured from a nearby carrier.

4.2 Construction 1 (Profile-Relative Wall Corpus)

The corpus type is

$$\text{WallCorpusEntry} = \text{MorphologyRecordBundle} \sqcup \text{WallContextRecord}.$$

The morphology-bundle type is

$$\text{MorphologyRecordBundle} = \text{StrictWallRecord} \sqcup \text{AnalogueMorphologyRecord}.$$

A morphology record bundle contains a finite collection of morphology atoms. A `StrictWallRecord` has an upstream-admitted primary wall field. An `AnalogueMorphologyRecord` is included by the Paper XI corpus construction without asserting upstream strict-wall admission. A wall context record carries a pre-wall reference, static boundary witness, trajectory diagnostic, or retired provenance entry and contains no morphology atom.

For an admissible datum $\mathfrak{D}_\kappa$, define

$$\begin{aligned} \text{RecordWall}_{P_W}(\mathfrak{D}_\kappa) &\in \text{StrictWallRecord}, \\ \text{RecordWall}_{P_W}(\mathfrak{D}_\kappa) &= (\text{Addr}, \text{Role}, \text{Scope}, \\ &\quad \text{Policy}, \text{Events}, \text{Evidence}). \end{aligned}$$

where **Addr** is the source address and **Events** is a finite collection of morphology atoms, ordered only when its declared context supplies an order. This is a record-producing construction. No functor arrow, naturality law, or weak-morphism transport is asserted.

Analogue morphology enters through a separate corpus-inclusion construction, denoted $\text{RecordAnalogueMorphology}_{P_W}$ when a name is needed. It does not consume an upstream-admitted wall datum and does not produce a **StrictWallRecord**.

For a morphology atom e in a morphology bundle, the atom record is

$$W_e = (\text{id}, \text{realization_kind}, \text{record_role}, \kappa_0, \text{primary_carrier_id}, \\ \text{context_field_keys}, \text{context_carrier_ids}, \text{atom}, \text{policy_refs}, \text{evidence_refs}).$$

The atom type is the disjoint union

$$\text{MorphologyAtom} = \text{TrajectoryEvent} \sqcup \text{LocusSample}.$$

The two summands are fully typed as

$$\begin{aligned} \text{TrajectoryEvent} = (\text{id}, \text{realization_kind}, \text{trajectory_ref}, \text{orientation}, \\ \text{event_interval}, \text{sampling_rule}, \text{primary_field}, \\ \text{context_fields}, \text{field_registrations}, \delta_e^\gamma, \\ \text{policy_refs}, \text{evidence_refs}), \end{aligned}$$

and

$$\begin{aligned} \text{LocusSample} = (\text{id}, \text{realization_kind}, \text{domain_context_ref}, \text{event_locus}, \text{primary_field}, \\ \text{context_fields}, \text{field_registrations}, \delta_e^{\text{loc}}, \text{probe_path_ref?}, \\ \text{probe_orientation?}, \text{order_semantics}, \\ \text{policy_refs}, \text{evidence_refs}). \end{aligned}$$

For an oriented trajectory event, the sparse change map is

$$\delta_e^\gamma = \left\{ q \mapsto (v_q^-, v_q^+, \text{change}_q) \right\}_{q \in K_e}.$$

Its record must declare a trajectory reference, orientation, left/right sampling rule, and event parameter or interval. The superscripts $-$ and $+$ refer to that declared orientation; they are not intrinsic labels on an ambient wall locus. The primary field of a **wall_event** must contain present, typed, and distinct endpoint values. **NOT_RECORDED** is therefore forbidden for either primary endpoint.

For a domain-level locus sample, the change data instead form an incident-stratum germ:

$$\delta_e^{\text{loc}} = \left\{ q \mapsto \{(C_\alpha, v_{q,\alpha})\}_{\alpha \in A_e} \right\}_{q \in K_e}.$$

Here the C_α are incident local strata. There need not be a unique before/after pair: a multiparameter bifurcation may have three or more incident strata. If a transverse probe path supplies two-sided samples, the record must retain its path and orientation and set **order_semantics** to **probe_relative**. Without such a probe it uses **intrinsic_none**; an oriented trajectory event uses **trajectory_oriented**.

The distinguished schema key $\kappa_0 = \text{primary_wall_field}$ names the primary field. In a **StrictWallRecord**, that field’s upstream admissibility establishes the wall atom. In an **AnalogueMorphologyRecord**, it anchors the recorded morphology but does not assert wall admission. Other keys in K_e are **context_field_keys**. Each context field must carry its own chart, comparison map, evidence reference, and every threshold, tolerance, or cutoff policy applicable to that field. Inapplicable policy slots are explicitly **not_applicable**. Thus the record has the typed form

$$\mathfrak{D}_{\kappa_0} \rightsquigarrow (\delta_{\kappa_0}, \{\delta_q^{\text{context}}\}_{q \in K_e \setminus \{\kappa_0\}}).$$

A context-field change is a co-observation only. It does not become a second strict wall atom without independent wall admission, or a second analogue atom without separate corpus inclusion. A constant context field does not disprove the primary wall. In particular, recording a Lie/Hall wall beside a word-field change does not identify the two carriers.

Each key q is a typed tuple

$$q = (f_q, n_q, c_q),$$

where f_q is its field family, n_q is its registered field name, and c_q is the convention map containing the applicable carrier, alphabet, depth, pair-scope, reduction, and policy parameters. Representative registrations are:

Field family	Registered field name	Convention parameters
operator	<code>operator.direct_support[Y]</code>	labelled alphabet Y
route	<code>route.support[Y,d=2]</code>	route length $d = 2$
word	<code>word.support[Y,d=2]</code>	word length $d = 2$
word	<code>word.depth_truncated[Y]</code>	cutoff 6
Lie-Hall	<code>lie.simple_commutator_support[X]</code>	registered Lie family X
Lie-Hall	<code>lie.depth_truncated[X,H]</code>	Hall convention H , cutoff 4
closure	<code>closure.observable_star.marked_corner_dimension[Y,i,j]</code>	off-diagonal pair scope
closure	<code>closure.observable_star.marked_corner_dimension[Y,i,i]</code>	Hilbert–Schmidt
closure	<code>closure.sector_star.marked_corner_type[Q,Y,i,j]</code>	scalar-complement reduction
spectral	<code>spectral.gap</code>	marked sectors Q and alphabet Y
proxy	<code>proxy.response_time[policy]</code>	declared normalization and gap policy
state	<code>state.first_hit_time[policy]</code>	declared response policy
		declared first-hit policy

Raw diagonal unital corners are not compared without a declared reduction convention. An unrecorded endpoint may carry the record-field presence marker **NOT_RECORDED** only on non-admitting context or reference fields; an absent capability is typed as unavailable. **NOT_RECORDED** is neither a Paper X IR result state nor a Paper XIII comparison state. Neither marker is numerical zero.

4.2.1 Directional Semantics

Every trajectory delta also carries the direction tuple

$$\text{Dir}_e(q) = (d_n(q), d_a(q), d_s(q)),$$

whose components have distinct meanings:

Component	Meaning
d_n	direction of the recorded numerical value
d_a	direction of accessibility under the field's declared order
d_s	event-relative meaning, such as gain, loss, repair, or terminalization

For example, an endpoint repair may decrease a depth or unreachable-pair count while increasing accessibility. Direction without this qualifier is not a valid wall coordinate.

Profile-Relative Wall Corpus

Paper XI separates strict and analogue morphology bundles from non-wall context records.

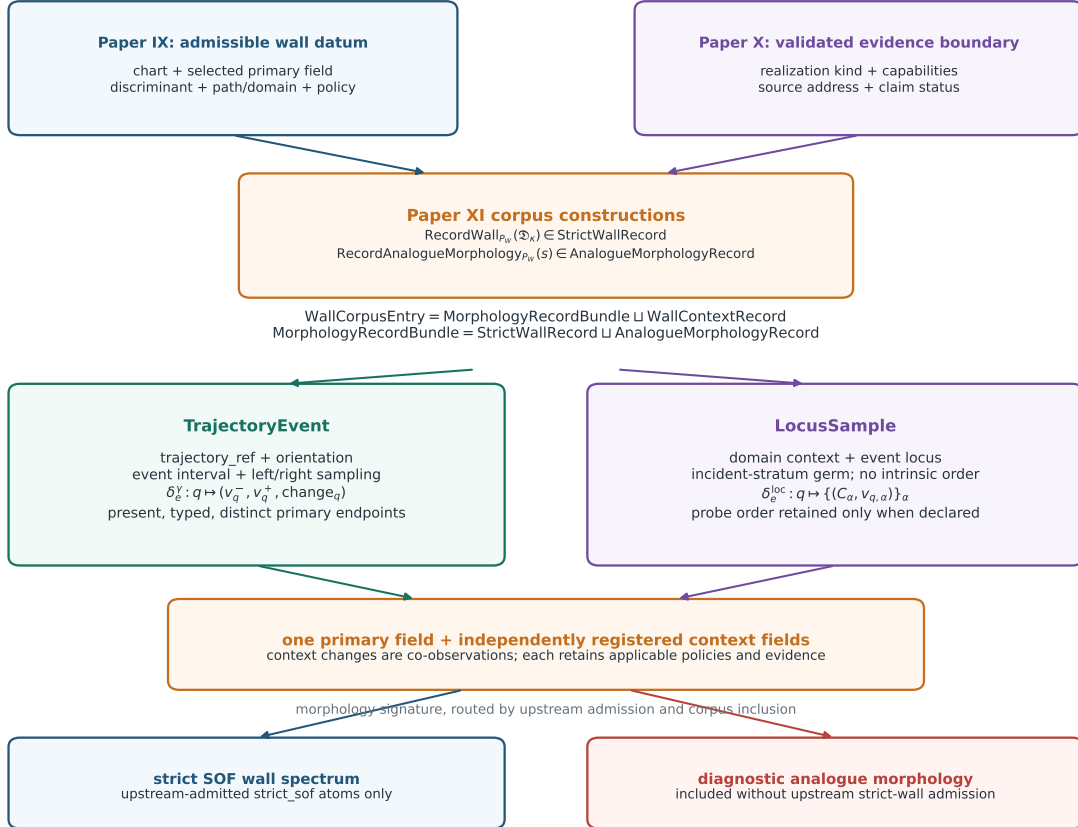


Figure 1. Profile-relative wall-record construction. Paper IX supplies wall-admissibility conditions and deformation data, while Paper X supplies capability and evidence guards. Paper XI records a discriminated morphology atom: an oriented trajectory event has a sparse before/after map, whereas a domain-level locus sample has an incident-stratum germ without intrinsic two-sided order. The primary field and context co-observations remain separately registered, and strict-SOF signatures remain separate from diagnostic-analogue morphology.

4.3 Independent Pair States

For a fixed-schema pair audit, the state is a bundle rather than a ladder:

$$S_{ij}(t) = (s_{\text{op}}, s_{\text{route}}, s_{\text{word}}, s_{\text{Lie}}, d_{\text{word}}^{(\leq m)}, d_{\text{Lie}}^{(\leq n)}, s_{\text{terminal}}),$$

with only declared components retained. Direct support, a word witness, and a Lie witness may coexist. Terminality is not a competing transport layer. Adjacent samples produce the sparse component change

$$\delta S_{ij}(t_k) = S_{ij}(t_{k-1}) \rightsquigarrow S_{ij}(t_k),$$

where the arrow denotes a recorded before/after transition, not subtraction in a common vector space.

4.4 Closure Fields

The three closure layers remain distinct:

$$A_Y^+, \quad A_Y^*, \quad A_{Q,Y}^*.$$

The record schema may include dimension or corner-dimension changes for the positive algebra. Observable-star and sector-star records may separately track marked corners, abstract algebra type, and concrete embedding. An unchanged abstract star-algebra type does not imply unchanged marked corners or concrete embedding.

Finite-dimensional C^* -algebras admit standard Wedderburn decomposition, but A_Y^+ is not automatically a C^* -algebra or semisimple. A positive-algebra field may use Wedderburn terminology only when a separate semisimplicity certificate is declared.

5 Wall Coordinate Profiles

5.1 Six Structured Coordinate Families

The derived coordinate profile is

$$\text{Prof}_W^{P_W}(e) = (\text{FieldFamilies}, \text{EventKinds}, \text{Regularity}, \text{PersistenceProfile}, \text{Geometry}, \text{Evidence}).$$

These families use the following controlled vocabulary:

Family	Values
Field families	operator, route, word, Lie-Hall, closure, spectral, state, graph, stochastic, proxy, or another declared typed family
Event kinds	any declared subset of support gain/loss, rank jump, first-hit change, collision, repair, terminalization, plateau onset/exit, response-order crossing, boundary hit
Regularity	smooth, stratified, piecewise smooth, discrete, stochastic, unknown
Persistence profile	any declared subset of transient, persistent, oscillatory, plateau, terminal, unresolved
Geometry	location: interior/boundary/endpoint; crossing: transverse/tangent/unresolved; codimension status: certified/unresolved
Evidence	Theorem, Computational Certificate, Computational Observation, Research Program

Event kinds and persistence entries need not be single-valued, and geometry is a structured object rather than one categorical coordinate. Censoring and sampling remain observation policies in P_W , not regularity types. These entries are wall coordinates or diagnostics, not automatically invariants. Crossing counts depend on sampling; lengths and response times depend on parameterization and policy; truncated depth depends on cutoff; spectral gaps depend on normalization; codimension depends on the ambient chart; and census prevalence depends on the selected sample.

5.2 Wall-Profile Relativity Principle

Principle (Wall-Profile Relativity). A source system does not determine a unique wall record or wall spectrum. The result is indexed by its typed realization, chart, selected field, parameter domain or trajectory, and Wall Profile.

Accordingly, morphology identity is defined without curation metadata:

$$\text{MorphSig}_W^{P_W}(e) = (\text{RealizationKind}(e), \text{Role}(e), \text{AtomKind}(e), \text{PrimaryFieldFamily}(e), \\ \text{PrimaryCarrierId}(e), \delta_e, \text{Prof}_W^{P_W}(e)).$$

where δ_e denotes δ_e^γ or δ_e^{loc} according to the atom type. A versioned curated view is the separate object

$$\text{CuratedSig}_W^{P_W, v}(e) = (\text{MorphSig}_W^{P_W}(e), \text{Curate}_v(e)).$$

Changing a curation rulebook may therefore change a curated view without changing the morphology identity. For a finite corpus $\mathcal{C}$ of `WallCorpusEntry` values, the profile-relative strict wall spectrum of $\mathcal{F}$ is

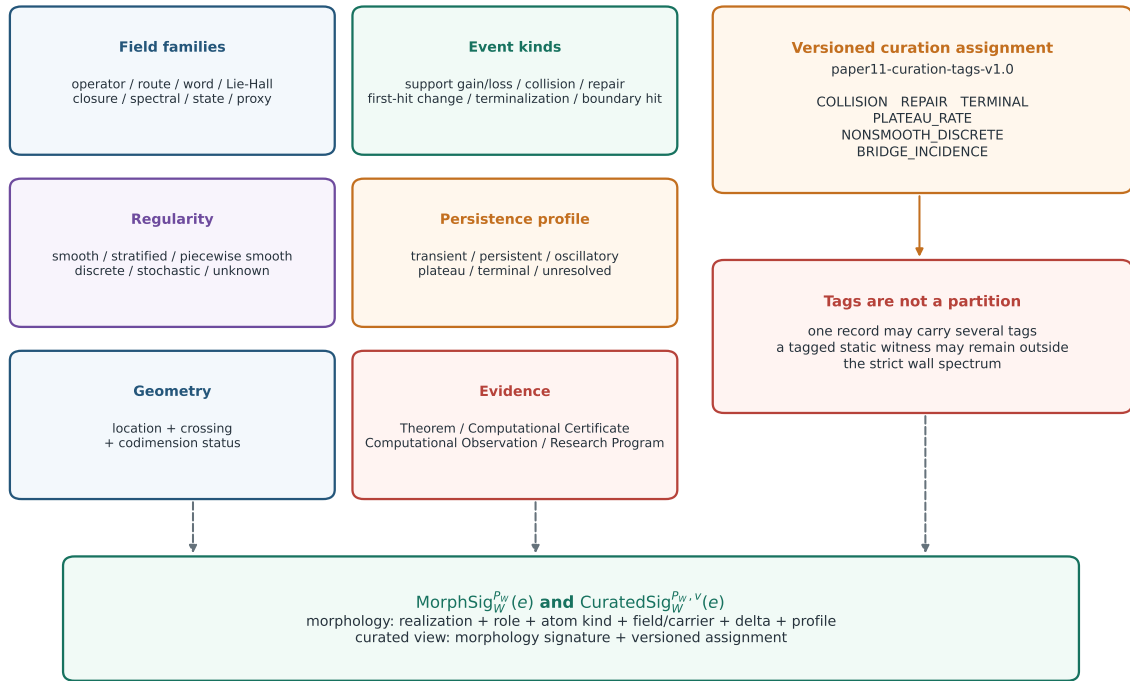
$$\text{Spec}_{W, \text{str}}^{P_W}(\mathcal{F}, \mathcal{C}; \chi) = \left\{ \left\{ \text{MorphSig}_W^{P_W}(e) : \begin{array}{l} e \in \mathcal{C}, \\ e \text{ is attached to } \mathcal{F}, \\ e \text{ is upstream-admitted} \\ \text{and is a strict-SOF morphology atom} \end{array} \right\} \right\}.$$

Corpus-included diagnostic analogues define a separate set $\text{Morph}_{W, \text{an}}^{P_W}(\mathcal{C}; \chi)$; they are not members of $\text{Spec}_{W, \text{str}}^{P_W}$. Record role alone therefore never promotes an analogue into the strict wall spectrum.

The context χ is either a trajectory context, including its orientation and sampling rule, or a domain context, including its incident-stratum convention and any probe. Different choices of sectorization, alphabet, closure, filtration, path or domain, threshold, cutoff, or profile may yield different spectra. This is a definition-level relativity statement, not a theorem that two profiles are equivalent or comparable. Cross-profile and cross-report alignment belongs to the comparison interface of Paper XIII.

Structured Wall Coordinates and Nonexclusive Curation

Morphology identity, observation policy, and versioned curation remain separate typed objects.



Sampling, censoring, threshold, normalization, and cutoff remain policies in P_W ; they are not morphology axes.

Figure 2. Structured wall coordinates and nonexclusive curation. The six coordinate families form a typed morphology profile; sampling, censoring, threshold, normalization, and cutoff remain policies in the Wall Profile. The versioned curation tags are multi-label assignments rather than partition classes, and neither coordinates nor tags promote a record into the strict wall spectrum.

6 Multi-Label Taxonomy

The six mechanism families are retained as full curation tags:

Curation tag	Typical use
Collision	spectral or observable branches meet under a declared collision policy
Repair	a previously absent or cutoff-unreached typed field witness, support state, or finite-filtration reachability state becomes present
Terminal	a terminal component, stopping region, or endpoint structure changes
Plateau / rate	plateau or response-order morphology is recorded
Nonsmooth / discrete	piecewise-smooth, rank-selection, discrete, or censored behavior occurs
Bridge / incidence	route, word, commutator, or image-kernel mechanism is implicated

The tags are not a partition and do not replace the structured coordinate families. One event may be simultaneously **REPAIR**, **NONSMOOTH_DISCRETE**, and **BRIDGE_INCIDENCE**. Conversely, a static **REPAIR**-tagged finding may remain a `static_boundary_witness` and therefore stay outside the wall spectrum. The slash-separated display names map to the corresponding underscore-separated keys in the machine-readable census.

Each assignment is a versioned object

$$\text{Curate}_v(e) = (\text{rulebook_version}, \text{assignment_source}, \text{tags}, \text{override_reason?}),$$

not an untracked list copied from a source letter label. The active rulebook is **paper11-curation-tags-v1.0**; it declares nonexclusive membership and records any typed override to an inherited assignment.

Letter names are not used for the active tags because they suggest a partition and conflict visually with Arnold’s A_k, D_k, E_k notation.

7 Local-Model Eligibility

For any proposed local-model family $\mathcal{M}$ and local branch B of a declared discriminant, define the general eligibility predicate

$$\text{Eligible}_{\mathcal{M}}(B; P_W) \in \{\text{yes}, \text{no}, \text{unresolved}\}.$$

The predicate asks whether the branch, evidence, and declared equivalence notion satisfy the hypotheses of $\mathcal{M}$; it does not assign a model. ADE terminology enters only as the smooth-singularity specialization

$$\text{Eligible}_{\text{ADE}}(B; P_W) := \text{Eligible}_{\mathcal{M}_{\text{ADE}}}(B; P_W).$$

ADE eligibility requires all of the following:

1. a smooth finite-dimensional ambient chart near the point;
2. a locally defined smooth discriminant branch;
3. fixed carrier, labels, ranks, and field conventions;
4. sufficient differentiability for the proposed normal-form comparison;
5. a declared local equivalence notion and ambient dimension;

6. source-addressed evidence for transversality, tangency, or the relevant jet data;
7. explicit isolation of, or accounting for, unrelated invariant blocks and symmetry-protected degeneracies.

The machine record retains the legacy Boolean under `compatibility.historical_census_eligible`. It does not reuse that field for `local_model_eligibility`. Its `status` is `no` only when at least one necessary condition is certified false; missing or unchecked evidence yields `unresolved`.

An affirmative value means only that an ADE comparison is mathematically meaningful. The local-model branch must first be identified:

Branch	Required treatment
smooth discriminant	smooth local equivalence and jet analysis
stratified	stratum and incidence data
piecewise smooth	side-specific smooth data and switching rule
discrete	combinatorial event description
algebraic incidence	ambient algebraic variety and rank conditions
stochastic	stochastic model together with its declared sampling, censoring, and uncertainty policy

These branches require distinct tools [16, 11, 13].

The branch-aware audit in Appendix A is a negative eligibility control: simultaneous small sorted eigenvalue gaps do not establish $A_2 \rightarrow A_1 + A_1$ adjacency.

8 Evidence Anchors

8.1 Smooth-candidate Spectral Branch

A constructed real-symmetric endpoint has two isolated adjacent pair-gap closures. Degenerate perturbation theory supports recording them as first-order pair-gap closure candidates under the declared perturbation audit, subject to the real-symmetric crossing boundary [20]. The independent branch-aware Rubik audit continues eigenbranches within the *cp*, *ep*, *co*, and *eo* invariant blocks and detects no tested $A_2 \rightarrow A_1 + A_1$ split candidate. Thus pair-gap morphology is retained while an unsupported ADE adjacency is rejected.

Claim status. Computational Certificate for the declared matrices, parameter slices, and branch-matching rule. No generic ADE classification is claimed.

8.2 CNOT Path-Admissibility Diagnostic

Let $U_C = U_{\text{CNOT}}$. Since $U_C = U_C^* = U_C^{-1}$, define its spectral projectors

$$P_- = \frac{I - U_C}{2}, \quad P_+ = I - P_-.$$

The previously used strength interpolation is the affine path

$$U_{\text{aff}}(s) = I + s(U_C - I) = P_+ + (1 - 2s)P_-.$$

It is singular at $s = 1/2$: its rank is 3, its determinant is zero, and no finite matrix logarithm exists. Away from that point, the samplewise scalar convention $\text{Arg}(-x) = \pi$ gives

$$L_{\text{aff}}(s) = \begin{cases} \log(1 - 2s)P_-, & 0 \leq s < 1/2, \\ (\log|1 - 2s| + i\pi)P_-, & 1/2 < s \leq 1. \end{cases}$$

Thus the skew-adjoint part is zero on the left component and $i\pi P_-$ on the right component. The corresponding $0/12 \rightarrow 6/12$ side difference is tied to a singular logarithm-domain crossing and a samplewise branch convention; it cannot be promoted to an ordinary Lie/Hall trajectory wall.

The control replaces the affine interpolation by the fractional-CNOT path

$$U_{\text{uni}}(s) = P_+ + e^{i\pi s} P_-, \quad L_{\text{uni}}(s) = i\pi s P_-.$$

Here $U_{\text{uni}}(s) = \exp L_{\text{uni}}(s)$ is unitary and invertible for every $s \in [0, 1]$, with $U_{\text{uni}}(0) = I$ and $U_{\text{uni}}(1) = U_C$. The H and S logarithms are held fixed using explicit anti-Hermitian branches. On the declared 21-point grid, the bounded support tuple

$$\left(N_{\text{direct,unreached}}, N_{\text{Lie,unreached}}^{(\leq 4)}, N_{\text{Lie,reached without direct}}^{(\leq 4)} \right)$$

equals $(8, 8, 0)$ at $s = 0$ and $(6, 0, 6)$ at every sampled $s > 0$. Hence the admissible control exhibits one-sided endpoint activation but no sampled interior transition near $s = 0.55$. The previously reported repair bracket $(0.50, 0.55]$ is withdrawn rather than repaired by reparameterization.

Claim status. The projector and path formulas are exact finite-dimensional identities. The thresholded 21-point support census is a Computational Observation. The retained record role is `trajectory_diagnostic`, with `wall_admission=not_admitted`; no physical Hamiltonian mechanism or interior repair wall is claimed.

8.3 Nonsmooth/Discrete Branch

In the graph edge-weight control, the typed state bundle is constant for $t < 1$ and changes only when the weakened edges vanish at the endpoint. The certificate records 6 pair events and 12 independent field changes at that step. This is a discrete endpoint wall record; it is not assigned a smooth codimension or ADE type.

Claim status. Computational Certificate for the declared 11-sample path.

The nested-percolation audit records the statistical field `word.unreached_pair_count_at_cutoff[Y,cutoff=6,aggregation=ensemble_mean]`. Its value is an ensemble mean of integer pair counts, not an individual truncated word depth. The largest adjacent sampled drop occurs in the declared probability sweep, but no threshold discriminant was fixed upstream. It is therefore a `trajectory_diagnostic`, not a strict wall event.

Claim status. Computational Observation for the seeded 32-member ensemble and cutoff 6.

9 Census Certificate

The typed census is generated from a SHA-256-pinned 28-row source census. Corpus inclusion, record roles, and upstream admission references are read from the versioned `results/wall_record_inclusion_ledger_v1.json` rather than inferred from source identifiers. This ledger is not a wall-admission authority.

Record role	Count
pre_wall_reference	2
retired_provenance	1
static_boundary_witness	5
trajectory_diagnostic	13
wall_event	5
wall_locus_sample	2

The independent realization-kind census contains 19 `strict_sof` and 9 `diagnostic_analogue` rows. Seven morphology record bundles contain seven morphology atoms: 5 strict bundles enter the strict wall spectrum and 2 analogue bundles enter only the analogue morphology set. In this snapshot every morphology bundle contains exactly one morphology atom, so bundle and atom counts coincide; the schema does not identify these notions. The seven atoms contain 10 registered atom-field entries: 8 trajectory-change entries and 2 locus-germ entries. Four entries are pair-scoped. These entry counts are not counts of changed ordered pairs inside an aggregate field.

In the compact table below, `CC` and `C0` abbreviate the controlled statuses Computational Certificate and Computational Observation, respectively.

Record	Corpus status	Primary field	Evidence
Rubik collision quotient	strict locus	<code>spectral.joint_sector_count</code>	CC
Rubik endpoint closures	strict event	<code>spectral.adjacent_gap</code>	CC
constructed route	strict locus	<code>route.support[Y,d=2]</code>	CC
incidence			
graph edge endpoint	strict event	<code>operator.direct_support[Y]</code>	CC
constructed GOE	strict event	<code>spectral.adjacent_gap</code>	CC
endpoint			
dynamic maze split	analogue morphology	<code>graph.component_count</code>	C0
GRN basin loss	analogue morphology	<code>state.terminal_component_count</code>	C0

Every trajectory event in this table has present, typed, and distinct primary endpoint values. The CNOT logarithm boundary, nested percolation sampled drop, and barrier first-hit proxy are `trajectory_diagnostic` records rather than admitted events. Among 27 active records, the regenerated nonexclusive curation assignments have 34 memberships:

Tag	Active records	Strict-wall records
<code>COLLISION</code>	4	3
<code>REPAIR</code>	9	0
<code>TERMINAL</code>	3	0
<code>PLATEAU_RATE</code>	4	0
<code>NONSMOOTH_DISCRETE</code>	7	1
<code>BRIDGE_INCIDENCE</code>	7	1

The zeros in the strict-wall column are intentional. Plateau/rate rows remain diagnostics, while included terminal morphology is presently analogue rather than strict. No coverage target requires every tag to appear in the strict wall spectrum.

This finite source-addressed census has Computational Certificate status. Its counts do not estimate population prevalence, prove tag independence, or establish taxonomy completeness.

Typed Census and CNOT Path-Admissibility Diagnostic

Current source-addressed records; no prevalence, wall bracket, or repair threshold is inferred.

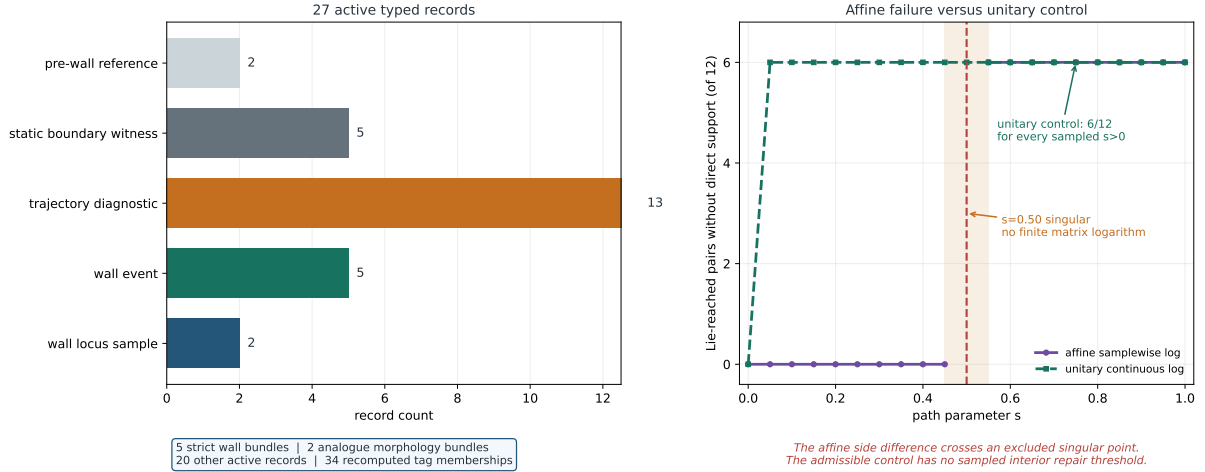


Figure 3. Typed census and CNOT path-admissibility diagnostic. The left panel shows the 27 active records by role and separately states the 5 strict wall bundles, 2 analogue morphology bundles, and 34 regenerated nonexclusive tag memberships. The right panel compares the singular affine interpolation with the unitary fractional-CNOT control. The affine path changes across an excluded logarithm-domain point; the latter has one sampled signature for every $s > 0$ and no interior repair threshold.

10 Discussion and Ownership

The logical center is one-record morphology: upstream-admitted strict wall records and separately included analogue morphology. Paper IX owns wall admission, and Paper X owns capability and evidence guards. Paper XI neither infers a wall from a static SOF nor fills missing upstream data.

The discriminated morphology-atom model resolves the orientation ambiguity: trajectory events carry declared before/after order, whereas locus samples carry incident-stratum germs. The realization/role/field-family split prevents analogue status from being mistaken for event function or mathematical carrier. Exact operator, route, word, Lie/Hall, closure, spectral, state, and proxy fields can coexist as separately registered observations without being promoted into one carrier or projected into a mandatory vector. The same principle rules out a mutually exclusive pair-status ladder: independent components may be simultaneously true, and the cutoff-unreached marker remains a finite-audit state.

The profile-relative terminology also explains why many quantities sometimes called “invariants” are only coordinates here. Their values remain useful, but only relative to the declared parameterization, cutoff, threshold, normalization, ambient chart, or sample. An invariance claim requires a separate equivalence relation and proof.

The evidence anchors align with this boundary. The spectral audit tests local smooth-model eligibility; the CNOT audit exposes a logarithm-domain boundary and uses an admissible unitary control to reject the reported interior threshold; and the graph endpoint exhibits admitted discrete morphology. The finite census then shows that one record may carry multiple mechanism tags while retaining one explicit record role.

The ownership boundary is equally important. Paper XII may present these records in a single report, and Paper XIII may align coordinates from two reports. Neither downstream operation changes a wall record or supplies missing upstream evidence.

11 Claim Status and Boundary

Definitions of record structure, coordinate families, record roles, curation tags, and eligibility predicates are not additional evidence levels. Reader-facing claims use exactly four levels:

Claim	Status
imported wall-pullback and calibrated response results	Theorem
typed census, sampled trajectory audit, independently validated	Computational
collision/incidence records, and branch-aware spectral audit	Certificate
CNOT domain diagnostic and bounded percolation, Kuramoto, GRN, neural, and application-specific records	Computational Observation
wall-record functoriality, complete taxonomy, stable metrics, and universal local models	Research Program

The theorem row remains owned by the cited upstream paper. Certificates are finite and source-addressed; observations remain bounded by their declared realizations and policies. Source artifacts retain their original ownership. This paper owns only the derived wall records, coordinate profiles, curation tags, eligibility decisions, and the finite census certificate.

11.1 Excluded Claims

This paper does not claim:

1. a functor from a static or deformation category to wall records;
2. that a static finding, plateau interval, response hierarchy, or proxy is automatically a wall;
3. that operator, route, word, and Lie/Hall support or depth are interchangeable;
4. that positive-word, observable-star, and sector-enriched star closures have the same wall coordinates;
5. that `UNREACHED_AT_CUTOFF` or any legacy numeric sentinel means infinity;
6. that the Paper VI rank 11/nullity 7 linearization is a nonlinear codimension theorem [6];
7. that sampled hit fractions are locus cardinalities or measures;
8. that wall coordinates are intrinsic invariants;
9. that the finite census is complete or prevalence-estimating;
10. that ADE types classify all SOF walls.

The exact response theorem and calibrated rate certificates remain owned by Paper IX. This paper may register their morphology as diagnostic context, but it does not turn a contraction-deficit ratio or response-time ordering into a wall-rate invariant.

12 Conclusion

This paper defines the morphology layer of the Sectorized Observable Framework program. In the strict branch, an admissible wall datum is converted by an ordinary, profile-relative construction into a bundle of oriented trajectory events or incident-stratum locus samples. The analogue branch uses a separate corpus-inclusion construction and makes no wall-admission claim. In both branches, the primary field and separately registered context fields retain their own carriers and policies. Realization kind, record role, and field family remain independent coordinates. Derived morphology uses structured field-family, event-kind, regularity, persistence, geometry, and evidence families. Mechanism labels are versioned nonexclusive curation assignments, and ADE comparison is a special case of a general local-model eligibility test.

The finite evidence supports this architecture without promoting it to a classification theorem. Five strict wall bundles enter the strict wall spectrum; two included analogue morphology bundles remain in a separate analogue morphology set, and static, pre-wall, diagnostic, and retired rows remain outside the strict wall spectrum. Spectral morphology, the excluded CNOT wall candidate with its admissible-path control, and the discrete graph endpoint demonstrate the intended separation of local-model eligibility, event morphology, diagnostic context, and evidence status.

13 Outlook

The open program is to define admissible equivalences and reparameterizations under which selected wall coordinates become invariant, to develop local models for stratified and nonsmooth branches, and to study transport between wall records only after an actual category of comparison data is defined.

Wall-record functoriality and naturality remain Research Programs. They must not be inferred from the strict categories of Paper VIII or from the record-only deformation interface of Paper IX.

Appendix A Branch-Aware Adjacency Falsification

The audit separates the Rubik representation into its invariant blocks and continues eigenbranches by maximum adjacent-sample eigenvector overlap. A candidate $A_2 \rightarrow A_1 + A_1$ split requires two isolated pair-gap minima at separated parameter values whose union involves three branches. Persistent degeneracies, endpoint-only closures, and cross-block equalities are rejected.

Block	Dimension	Endpoint cluster sizes	Tested triples	Split candidates
<i>cp</i>	64	8, 24, 24, 8	0	0
<i>ep</i>	144	12, 36, 24, 36, 36	0	0
<i>co</i>	8	3, 3	2	0
<i>eo</i>	12	3, 3, 3	0	0

No branch-aware split candidate is detected on the tested diagonal and asymmetric slices. A future positive candidate would still require local normal-form and versality analysis.

Appendix B Sampled Typed-State Trajectories

The controlled trajectory artifact declares four independent typed fields:

$$D_W^{(6)}[Y] := D_{\text{word}}^{(\leq 6)}[Y], \quad R_2^L[X] := R_2^{\text{Lie}}[X],$$

where the abbreviated symbols are used only in the following compact table.

Carrier	Typed field	Convention
operator	$R_1[Y]$	labelled alphabet Y
word	$W_2[Y]$	exact word length 2
word	$D_W^{(6)}[Y]$	truncated word depth at cutoff 6
Lie-Hall	$R_2^L[X]$	simple-commutator support on the independently registered Lie family X

Every adjacent pair event retains all changed fields. The generated summary is:

Control	Pair events	Field changes	Changed pairs	Change steps
GridWorld obstacle path	236	314	174	1, 2
SIR β sweep	4	10	4	1
Graph edge-weight endpoint	6	12	6	10

The counts are sampled-path coordinates. They do not establish continuous-time wall flow or ambient codimension.

Appendix C Computational Artifacts

The following repository artifacts support the typed census, included morphology records, finite robustness checks, and bounded observations used above. Unless another base directory is stated, short paths are relative to `experiments/paper11/`.

C.1 Census and Corpus Inclusion

Artifact	Role	Short path
C1	typed census producer and validator	<code>validation/typed_wall_record_census.py</code>
C2	versioned corpus-inclusion and upstream-admission reference ledger producer and result	<code>validation/build_wall_record_inclusion_ledger.py</code> ; <code>results/wall_record_inclusion_ledger_v1.json</code>
C3	generated 28-row typed census and ownership summary	<code>results/wall_record_census_typed_v3.json</code> ; <code>results/wall_record_census_typed_v3.md</code>

C1 verifies the SHA-256-pinned source census, the inclusion ledger, and every ledger evidence contract before generating C3. C2 separates upstream wall admission from Paper XI corpus inclusion and binds each included entry to its realization kind, record role, primary field, evidence schema, artifact SHA-256, producer SHA-256, and reader-facing claim status. The resulting census is a Computational Certificate for this finite source-addressed curation; it is not a completeness or prevalence theorem.

C.2 Wall Records and Bounded Audits

Artifact	Role	Short path
C4	typed pair-state trajectory producer and result	wall_trajectory.py; results/wall_trajectory.json
C5	CNOT affine-path failure and unitary-path control producer and record	cnot_logarithm_boundary.py; results/cnot_logarithm_boundary_v1.json
C6a	Rubik spectral-endpoint producer and record	spectral_ade_collision.py; results/rubik_spectral_endpoint_v1.json
C6b	constructed GOE endpoint producer and record	validation/degenerate_endpoint_collision.py; results/constructed_goe_endpoint_v1.json
C6c	Rubik collision-quotient producer, result, independent validator, and certificate	validation/produce_collision_quotient_result.py; results/rubik_collision_quotient_result_v1.json; validation/validate_collision_quotient_result.py; results/rubik_collision_quotient_validation_v1.json
C6d	route-incidence producer, result, independent validator, and certificate	validation/produce_route_incidence_result.py; results/route_incidence_result_v1.json; validation/validate_route_incidence_result.py; results/route_incidence_validation_v1.json
C7	nested-percolation producer and trajectory diagnostic	validation/percolation_diagnostic.py; results/percolation_diagnostic_v1.json
C8	GRN terminal-basin producer and wall record	validation/grn_toggle_wall.py; results/grn_terminal_basin_loss_v1.json
C9	bounded profile-robustness producer and record	validation/wall_robustness_audit.py; results/wall_robustness_v1.json
C10	branch-aware A_n falsification producer and results	an_adjacency.py; results/an_adjacency.json; results/an_adjacency.md
C11	bounded Markov producer and source-hashed observation	cross_species_wall_audit.py; results/markov_boundary.observation.json
C12	bounded Kuramoto producer and source-hashed observation	validation/kuramoto_wall.py; results/kuramoto_freezing.observation.json

From the repository root, run an executable artifact as `python experiments/paper11/<short path>`. C4–C10 support only the finite records and claim boundaries stated in their result artifacts. In particular, C10 is a negative adjacency control and does not establish a positive ADE classification. C5 and C7 are Computational Observations and do not enter the strict wall spectrum. C11 and C12 are Computational Observations cached through `experiments/observation.py`; stale source hashes block reuse, and the cached records do not enter the strict wall spectrum.

Imported evidence remains owned by its source manuscript and is bound through C2 and C3 rather than duplicated here. Paper XII may later serialize or present the resulting Paper XI records but supplies no upstream wall evidence here. A downstream Paper XII artifact retained as analogue corpus provenance does not constitute upstream wall admission. All listed artifacts are available in the [RIME repository](#).

References

- [1] Vladimir I. Arnold. *Catastrophe Theory*. Springer, 3 edition, 1992.
- [2] Vladimir I. Arnold, Sabir M. Gusein-Zade, and Alexander N. Varchenko. *Singularities of Differentiable Maps, Volume I*. Birkhäuser, 1985.

- [3] WuJun Chen. Capability-aware compilation for sectorized observable frameworks: Typed admission and cross-species registry evidence. *Preprint*, 2026. Paper X of the RIME program, version 2.0. DOI: <https://doi.org/10.5281/zenodo.21768257>.
- [4] WuJun Chen. Comparison geometry of SOF reports: Alignment, fixed-fiber pseudometrics, and word/lie transport separation. *Preprint*, 2026. Paper XIII of the RIME program. DOI: <https://doi.org/10.5281/zenodo.21449512>.
- [5] WuJun Chen. Incidence geometry of projected operator composition: Rank protection, image–kernel alignment, and promotion limits. *Preprint*, 2026. Paper VII of the RIME program, version 2.0. DOI: <https://doi.org/10.5281/zenodo.21634538>.
- [6] WuJun Chen. Linearized commutativity geometry on the generator-set moduli space: Full-matrix jacobian certificates, normality gates, and typed spectral registrations. *Preprint*, 2026. Paper VI of the RIME program, version 2.0. DOI: <https://doi.org/10.5281/zenodo.21634256>.
- [7] WuJun Chen. Observable dynamics of sectorized observable frameworks: Typed deformations, observable trajectories, and wall pullbacks. *Preprint*, 2026. Paper IX of the RIME program, version 2.0. DOI: <https://doi.org/10.5281/zenodo.21713306>.
- [8] WuJun Chen. Sectorized observable framework: A typed static object language for sectorized observables. *Preprint*, 2026. Paper VIII of the RIME program, version 2.0. DOI: <https://doi.org/10.5281/zenodo.21700863>.
- [9] WuJun Chen. SOF diagnostic protocol: A report specification for sectorized observable analysis. *Preprint*, 2026. Paper XII of the RIME program. DOI: <https://doi.org/10.5281/zenodo.21387462>.
- [10] Fan R. K. Chung. *Spectral Graph Theory*. American Mathematical Society, 1997.
- [11] Mario di Bernardo, Chris J. Budd, Alan R. Champneys, and Piotr Kowalczyk. *Piecewise-Smooth Dynamical Systems: Theory and Applications*. Springer, 2008.
- [12] Paul Erdős and Alfréd Rényi. On the evolution of random graphs. *Publications of the Mathematical Institute of the Hungarian Academy of Sciences*, 5:17–61, 1960.
- [13] Aleksei F. Filippov. *Differential Equations with Discontinuous Righthand Sides*. Kluwer Academic Publishers, 1988.
- [14] Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.
- [15] Martin Golubitsky and Victor Guillemin. *Stable Mappings and Their Singularities*, volume 14 of *Graduate Texts in Mathematics*. Springer, 1973.
- [16] Mark Goresky and Robert MacPherson. *Stratified Morse Theory*. Springer, 1988.
- [17] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 2 edition, 1995.
- [18] Yoshiki Kuramoto. Self-entrainment of a population of coupled non-linear oscillators. In *International Symposium on Mathematical Problems in Theoretical Physics*, volume 39 of *Lecture Notes in Physics*, pages 420–422. Springer, 1975.
- [19] Eugene Seneta. *Non-Negative Matrices and Markov Chains*. Springer, revised edition, 2006.
- [20] John von Neumann and Eugene P. Wigner. Über das verhalten von eigenwerten bei adiabatischen prozessen. *Physikalische Zeitschrift*, 30:467–470, 1929.